

MIN-MAX ± 1 QUADRATIC FORM

MathOverflow 413935 · existence of the limit is OPEN

STATUS

$\lim_{n \rightarrow \infty} \alpha_n$ is not proved to exist

Existence is the question. The value $1/2$ is unproved.

THE QUESTION

$$L = \lim_{n \rightarrow \infty} n^{-3/2} \min_{a_{ij} = \pm 1} \max_{x_i = \pm 1} \left| \sum_{i < j} a_{ij} x_i x_j \right|$$

Write $\alpha_n = m_n/n^{3/2}$. Does $\lim \alpha_n$ exist?

REVIEWED SANDWICH

1. $B \leq \liminf \alpha_n \leq \limsup \alpha_n \leq 1/2$,
with $B = 0.3258407554... > 13/40$ (paired polynomial-field, 2026-09-17).
2. Upper bound: Paley conference matrices, $\Phi(C) \leq \frac{1}{2}n\sqrt{n-1}$,
along $n = q + 1$ for primes $q \equiv 1 \pmod{4}$ (ratio-dense).
3. Elementary random signing: $m_n \leq \sqrt{\log 2} n^{3/2}$.
Consecutive gaps of α_n vanish; that does not force a limit.

REMARK

A Paley gap-2 argument for $L = 1/2$ is not a reviewed completion.
E(1) / residual (ii) are optional routes, not acceptance criteria.

WHAT IS PROVED / WHAT IS OPEN

two-ray criterion · Paley-skew shield · residue (6.20)

① TWO-RAY CRITERION (sufficient, not necessary)

If $H(n) = m_n^{2/3}$ obeys Dini-summable doubling and tripling,

$H(2n) \leq 2H(n) + 2n\eta(n)$ and $H(3n) \leq 3H(n) + 3n\eta(n)$, then α_n converges.

Neither ray is proved. Identifying the limit is not required.

② MULTIPLIER TWO — residue (6.20)

A balanced Paley-skew R shields every pair except Hamming-central,

high joint-energy pairs. Uncoupled $|x^T R y|$ budgets cannot close the diamond.

Exact minimizers at $n = 5, 6, 8$: residue is majority, Paley- R misses $\Theta(n^{3/2})$.

③ MULTIPLIER THREE / 1:2 SPLIT

Tetrahedral tripling frame and the 1 : 2 reduction are proved as identities.

The remaining tripling diamond and residuals (6.42)--(6.43) are open.

A cross-order pressure comparison is also open; B is not a limit value.

CONCLUSION

The original limit is OPEN. github.com/luckyseoul/quadratic-minmax-limit